Geophys. J. Int. (1999) **138,** 857–870

Palaeomagnetic constraints on the age of deformation of the Sierras Australes thrust and fold belt, Argentina

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Accepted 1999 May 10. Received 1999 April 15; in original form 1998 October 27

SUMMARY

A palaeomagnetic study has been carried out on late Palaeozoic rocks exposed in the Sierras Australes thrust and fold belt of Buenos Aires province (Argentina), in the early Permian red sandstones and clay siltstones of the Tunas Formation. The sections sampled are exposed in the eastern parts of the belt, in Sierra de las Tunas (north) and Sierra de Pillahuincó (south). More than 300 specimens were collected from 25 sites, in three localities with different structural attitudes. Demagnetization at high temperatures isolated a characteristic remanence at 20 sites. All the localities have a reverse characteristic remanence, suggesting that the magnetization was acquired during the Kiaman interval. Stepwise tectonic tilt correction suggests that the Tunas Formation in these localities acquired its magnetization during folding in early Permian times. Palaeomagnetic poles were computed for each locality based on partial tilt-corrected remanence directions. Taking into account the fact that these localities are close to one another and that the rocks are all of reverse polarity, a group syntectonic palaeomagnetic pole called Tunas was calculated: longitude: 13.9°E, latitude: 63.0°S; $A_{95} = 5.4°$, $K = 39.7$, $N = 19$. This pole is consistent with previously calculated poles from South America assigned to the early Permian. In age it corresponds to the early Permian San Rafaelic tectonic phase of the Sierras Australes. Independent geological evidence indicates that the Tunas Formation underwent syndepositional deformation. We conclude that the Tunas Formation was deposited, deformed and remagnetized, all during the early Permian.

Key words: Argentina, Gondwana, palaeomagnetism, Palaeozoic, Sierras Australes, South America, tectonics.

INTRODUCTION

The Sierras Australes are considered to be the South American counterpart of the Cape Fold Belt of South Africa. They trend approximately NW–SE (Fig. 1) in the southwest part of Buenos Aires province, between 37° and 39°S, and 61° and 63°W.

The geological features of the Sierras Australes have attracted the attention of many geologists since the first decades of this century. Traditionally, the deformation in the area (Fig. 1) has been explained with only one tectonic phase, assigned to the late Permian–Triassic interval, according to stratigraphic evidence and correlation with the Cape basin in South Africa (Harrington 1947; Varela 1978; Japas 1989; Von Gosen *et al.* 1990; and Rossello *et al.* 1997). Until now, however, there has been no direct evidence for the age of deformation. According to some investigators, there may have been tectonic phases in addition to the Permo–Triassic phase mentioned above. An early phase of deformation is thought to have occurred in the late Devonian–early Carboniferous interval, related to the Chañica phase (Massabie & Rossello 1984; Ramos 1984, 1988; López Gamundi & Rossello 1993; Tomezzoli 1997). Our interpretation of the Permo–Triassic phase is that it occurred in the early Permian. The area may also have been gently deformed during the formation and evolution of the Colorado basin (in the Late Mesozoic), which lies to the south, and during the Andean orogeny in the late Cenozoic (Quattrocchio *et al.* 1995; Pereyra 1996; Tomezzoli 1997), as suggested by evidence of uplift and reactivation of pre-existing faults. However the degree of this younger deformation is slight.

According to Ramos (1984, 1988), along the mutual northern boundary of Patagonia and southern margin of Gondwana, an ocean was consumed by subduction beneath Patagonia ending with collision of the continental block in the mid-Palaeozoic. As a consequence of this collision process, deformation in the

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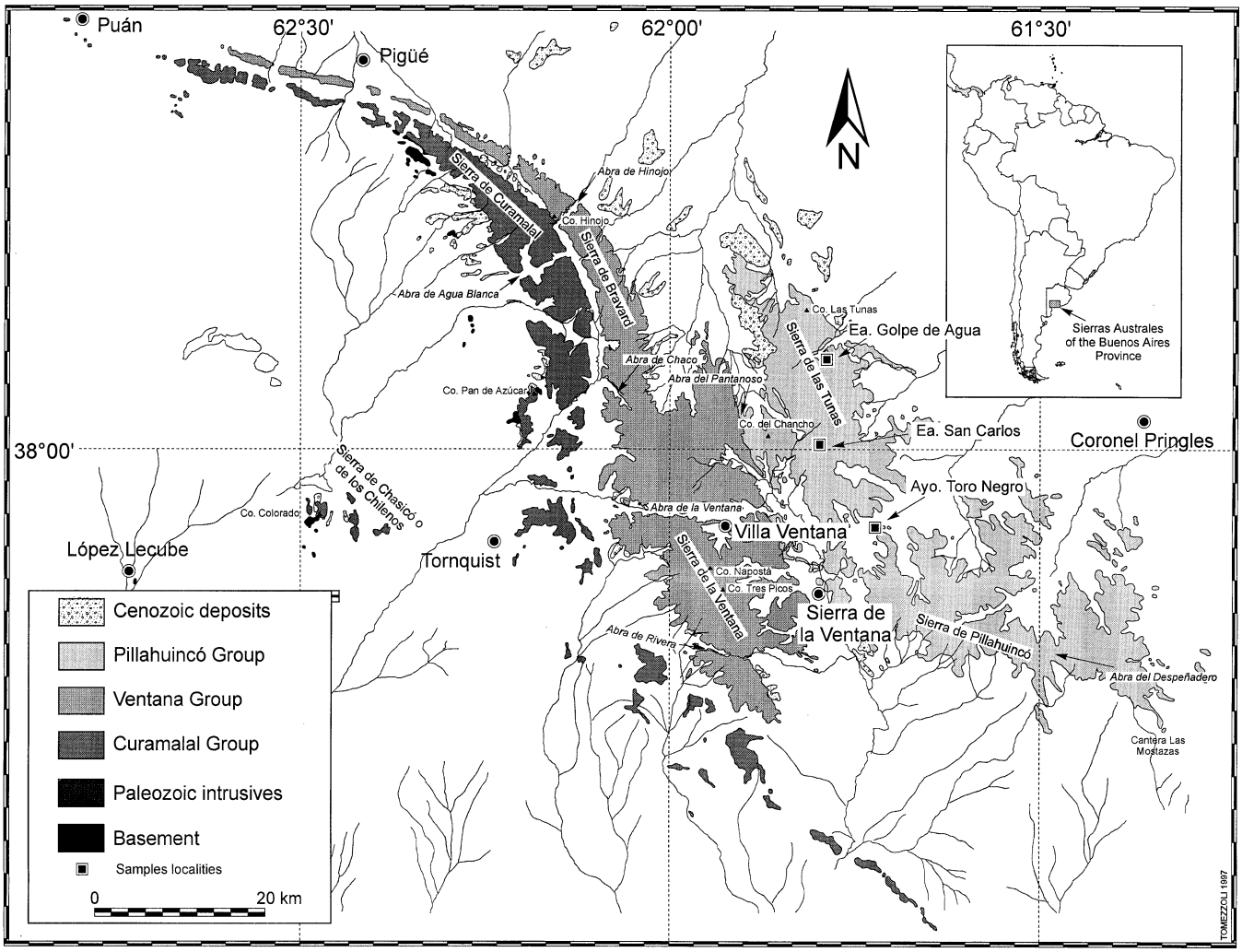


Figure 1. Geological map of the Sierras Australes exposed in the southwest of Buenos Aires province, Argentina. Squares indicate palaeomagnetic sampling localities.

Sierras Australes may have started in the mid-Palaeozoic, continuing to the early Permian (this work). Palaeomagnetic data that are available for Patagonia (Sierra Grande and Tepuel palaeomagnetic poles, Rapalini & Vilas 1991 and Rapalini *et al.* 1994, respectively) yield palaeolatitudes consistent with those expected from the South America poles from Devonian times onwards. The lack of significant latitudinal displacements of Patagonia with respect to Gondwana implies that accretion of Patagonia to Gondwana must have occurred during or before Devonian time.

Palaeomagnetism can help in the evaluation of the palaeogeographic and geodynamic evolution of the area. The only previous palaeomagnetic study in this area was undertaken by Creer *et al.* (1970), who obtained a palaeomagnetic pole for the Tunas and Bonete formations from the analysis of 10 samples (26 specimens). The results were interpreted as representing reverse pretectonic magnetization. The data, however, do not meet modern reliability requirements. The main objective of the present palaeomagnetic study is to obtain detailed palaeomagnetic evidence to constrain the palaeogeographic and tectonic evolution of the Sierras Australes of Buenos Aires province.

LOCAL GEOLOGY AND SAMPLING

The Sierras Australes is a fold belt of NW trend and with NE vergence. The range can be divided into two main orographic units (Fig. 1), with an important difference in the style of the deformation. In the western sector are the Curamalal, Bravard and Ventana ranges (Fig. 1), highly deformed, with up to eight levels of folding (Harrington 1947) and important thrusts (Tomezzoli & Cristallini 1998). In the eastern sector are the Tunas and Pillahuincó ranges (Fig. 1), with much gentler folding. These folds continue to the east in the subsurface, into the Interserrana or Claromecó basin (Llambías & Prozzi 1975; Tomezzoli & Vilas 1997).

The Tunas Formation (Harrington 1947) is widely exposed in the eastern sector. It is the youngest unit of the Carboniferous–Permian Pillahuincó Group (Furque 1973), which from base to top is made up of the Sauce Grande, Piedra Azul, Bonete and Tunas formations. The palaeoflora and palaeofauna associations of the Bonete and Tunas formations indicate a Permian age for these rocks. The *Eurydesma* fauna in the underlying Bonete Formation suggests a late Asselian to early

Sakmarian age (early Permian, Harrington 1947) and the *Glossopteris* zone according to Archangelsky & Cúneo (1984) suggests a Sakmarian to Artinskian age.

Sampling was restricted to the Tunas Formation, which is composed of over 1200 m of silicified fine pale green sandstones and lithic feldsarenites (Harrington 1947; Andreis *et al.* 1979; Andreis & Cladera 1992; López Gamundi 1996) interbedded with red clay siltstones and shale. Some beds have red spots which are interpreted as haematitic spots of diagenetic origin. According to Andreis *et al.* (1979), the red tones are characteristic of the Tunas Formation shale and fine sandstones (Fig. 1). These authors identified haematite by X-ray diffractometry and in thin sections. Some vitric tuff interbeds have been described in the upper half of the unit by Iñiguez *et al.* (1988), indicating contemporaneous volcanism close to the Tunas basin.

At least four hand samples or six drill cores were collected at each site (Fig. 1), representing different structural attitudes. The samples were orientated in the field using magnetic or sun compasses. Two standard specimens (2.2 cm long) were cut from each core.

PALAEOMAGNETIC STUDY

Measurements of natural remanent magnetization (NRM) were made using a DC Squid cryogenic magnetometer (2G model 750R). Pilot samples were submitted to stepwise alternating field (AF) demagnetization in steps of 2, 4, 6, 8, 10, 15, 20, 25, 30, 40, 50, 60, 80 up to 140 mT. This method proved ineffective in demagnetizing the specimens owing to the high coercivities of the remanence. Thermal demagnetization was more successful, and was applied in at least 15 steps, with intervals of 100°, 50°, 30°, 20°, 10° and 5 °C up to temperatures of 680 °C to 730 °C, using the Schonstedt Instrument Co. TSD-1 oven. Bulk susceptibility was measured after each step, in order to monitor possible chemical changes resulting from heating, with a RMSH-III TATA (Institute of Fundamental Research, Bombay, India) susceptibility meter.

Demagnetization results were plotted as orthogonal vector diagrams (Zijderveld 1967) and also as equal-area stereographic projections. Palaeomagnetic directions were determined using principal component analysis (Kirschvink 1980) or using the remagnetization planes technique (Halls 1976). The final mean site directions were analysed based on Fisher (1953) statistics, or, in the case of combined directional data and remagnetization circles, the method of McFadden & McElhinny (1988). All processing was carried out using the following software: CRIOIAPD (Cristallini 1995), IAPD (Torsvik 1990), CIRDI (Mena 1994), fold test (McFadden 1990b) and MAG88 (Oviedo 1989).

Palaeomagnetic results from the following localities are presented in this work: estancia San Carlos (38°S, 61°47′W), Golpe de Agua (formerly estancia Peñaflor, 37°55′S, 61°48.8′W), located in the Sierra de las Tunas, and arroyo Toro Negro (38°3.5′S, 61°42′W), between the Sierra de las Tunas and the Sierra de Pillahuincó (Fig. 1).

ESTANCIA SAN CARLOS

Oriented block samples were collected at 12 sites located on several limbs of a folded sequence (Fig. 2), with each site representing a different chronostratigraphic level. Two or more specimens were obtained from each block (146 specimens in total) and subjected to stepwise thermal demagnetization (Fig. 3). NRM intensities range between 1 and 10 mA m^{-1}. All samples showed similar behaviour during demagnetization (Fig. 3). Most of them showed a single component with very high unblocking temperature (up to 700–730 °C). They either have a linear decay towards the origin (Fig. 3a) or an abrupt decay of a very stable remanence (Figs 3b and c). In some samples, the demagnetization paths were curved, indicating the simultaneous removal of two distinct magnetic components with overlapping unblocking temperatures. Demagnetization circles were used to characterize these remanences (Fig. 3d). Two hand samples from site CT8 were rejected because of erratic behaviour. Site CT1 was excluded from the computation of the mean because its α_{95} value is greater than 15°.

The high-temperature characteristic remanence directions from the Tunas Formation in the estancia San Carlos were grouped as Population 1 (Table 1), having positive (downwards) inclinations and very good within-site consistency ($\alpha_{95} < 15°$ and $k > 20$). Unblocking temperatures between 630 °C and 730 °C clearly show that this magnetization is carried by haematite. In approximately 30 per cent of the specimens, a second, low-temperature component was isolated between 350 °C and 550 °C. This component has negative inclinations (*in situ*) and a tendency to be northward-directed (close to declination 0°). These component directions were grouped as Population 2.

Analysis

The *in situ* mean direction for Population 1 for estancia San Carlos (Fig. 4a), from 11 sites ($N = 11$), is Declination = 140.2°, Inclination = 56.9°, $\alpha_{95} = 8.0°$ and $k = 33.9$. After full 100 per cent bedding correction the mean direction is Declination = 149.9°, Inclination = 49.4°, $\alpha_{95} = 10.3°$ and $k = 20.7$ (Fig. 4b). Visual inspection of the distribution of the mean site directions before and after structural correction clearly indicates that the magnetization is not pretectonic (Fig. 4).

Application of stepwise structural correction (Figs 4c and 5) shows a clear improvement in the grouping of directions at around 30 per cent unfolding: Declination = 141.6°,

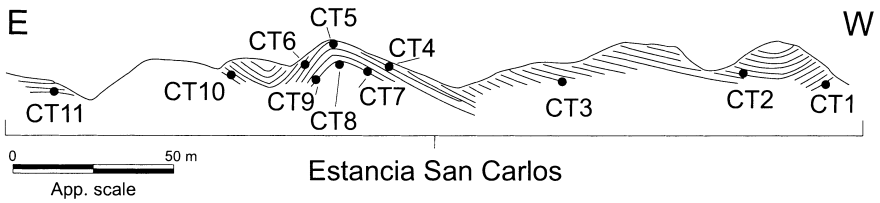


Figure 2. Schematic structural profile of the Tunas Formation at locality 1, estancia San Carlos, showing the positions of the sampling sites.

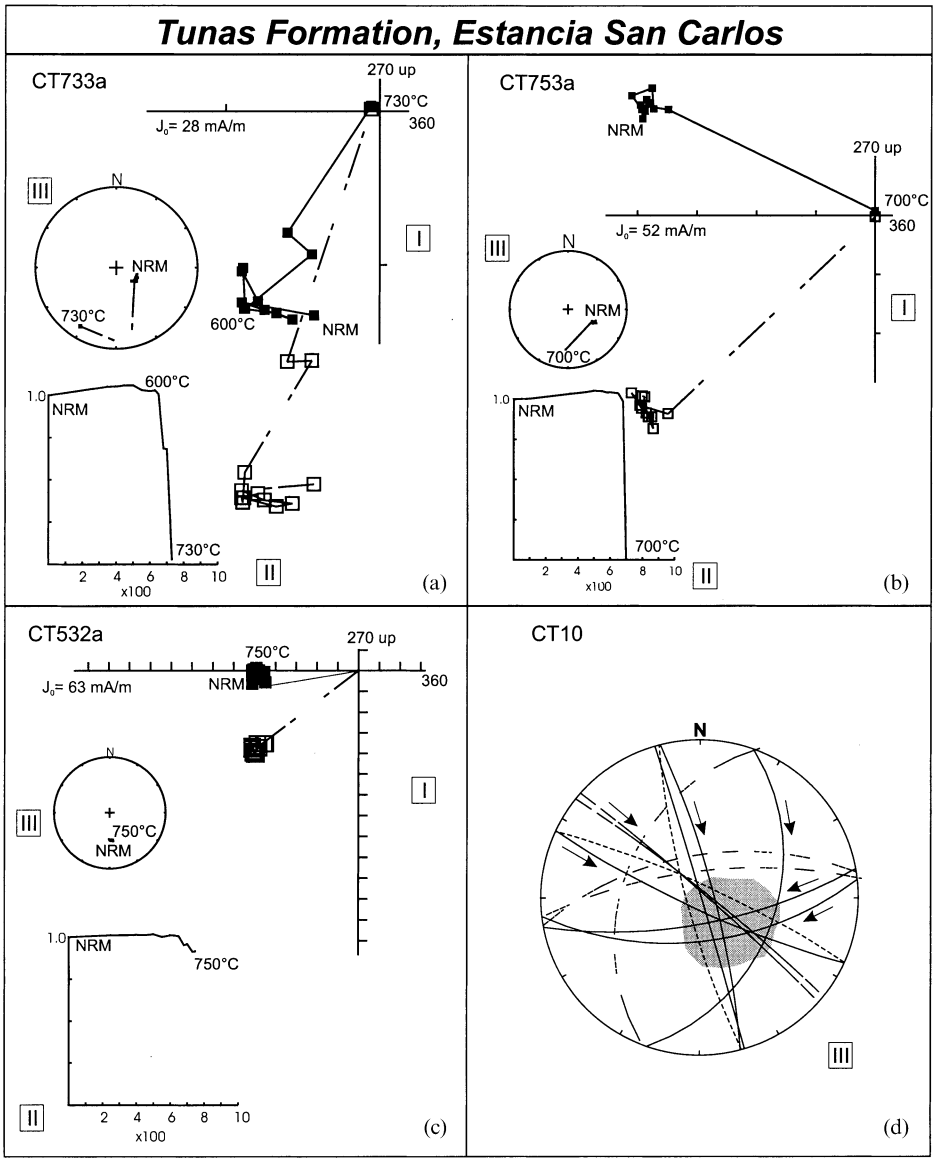


Figure 3. Representative thermal demagnetization results of some specimens from the Tunas Formation, in the estancia San Carlos (location shown in Fig. 1). I: Zijderveld diagrams. Open (filled) squares indicate representation in the vertical (horizontal) plane, in geographical coordinates. II: Demagnetization curves. III: Projection of an equal-area net. For (d), remagnetization circles are shown. The shaded area indicates the location of the mean direction.

Inclination = 58.7°, $\alpha_{95} = 4.5°$; $k = 131.9$, $N = 9$ and a minimum correlation ($\xi_{\text{Minimum}} = 0.048$) at 32 per cent according to definition 1 of McFadden's (1990b) fold test. This result suggests that the magnetization of the Tunas Formation in the estancia San Carlos was acquired during the folding of the sequence. Statistically, however, it is not possible to reject an *in situ* (post-tectonic) magnetization at the 99 per cent confidence level. As suggested by McFadden (1990a), CT5 and CT8 were not included in the calculations because they are located in the fold hinge. However, inclusion of these sites does not significantly alter the mean direction.

The lower-temperature Population 2 *in situ* mean direction is similar to the present geomagnetic field direction and is interpreted as a secondary magnetization acquired during recent times.

ESTANCIA GOLPE DE AGUA

This locality (formerly called estancia Peñaflor) is located northeast of estancia San Carlos (Fig. 1). Orientated block samples were collected at eight sites, with at least four hand samples at each one. Each site represents a different chronostratigraphic level and has a different structural position (Fig. 6). In this locality, well-preserved plant fossils (Lycophytes) are common.

Three specimens from each block (a total of 103 specimens) were subjected to stepwise thermal demagnetization. NRM intensities range from 1 to 60 mA m^{-1}. Similar demagnetization behaviour was observed in all sites, as in estancia San Carlos. Most of the samples showed high-stability components up to 680 °C, with discrete unblocking temperatures between

Table 1. Mean high-temperature characteristic directions in the Tunas Formation in estancia San Carlos. *N*/*n*: number of processed specimens/number of specimens used in the calculation of the mean; (*n*) number of great circles. Dec.: Declination (°); In.: Inclination (°); α_{95}(°) = semiangle of the 95 per cent confidence cone; *k*: Fisher statistical parameter (Fisher 1953). Bedding: strike and dip (90° cw from given strike).

Means of Population 1

	in situ						with structural correction							
							100 per cent				30 per cent			
Site	N/*n*	Dec.°	In.°	α_{95}°	*k*	Bedding	Dec.°	In.°	α_{95}°	*k*	Dec.°	In.°	α_{95}°	*k*
CT1*	9/4(3)	186	70	15.5	48	312/24	–	–	–	–	–	–	–	–
CT2	10/10	138	51	5	96	151/22	165	50	5	96	146	52	5	96
CT3	13/13	125	57	5	66	132/9	139	57	5	66	129	57	5	66
CT4	12/12	123	61	6	48	115/30	161	47	6	48	138	59	6	48
CT5	8/8	171	43	9.5	35	95/12	173	31	9.5	35	–	–	–	–
CT6	12/10	161	61	10	23	330/31	112.5	53	10	23	144	61	10	23
CT7	11/11	122	52.5	9	28	116/26	150	43	9	28	132	50	9	28
CT8	10/8	128	33.5	10	46	78/12	132	24	10	46	–	–	–	–
CT9	12/12(7)	149	69	7	44	329/18	110	63	7	44	135	68	7	44
CT10	14/8(8)	138	57	10	35	130/18	162	51	10	35	146	56	10	35
CT11	10/10	148	63	4	169	123/13	166	56	4	169	154.5	61	4	169
CT12	11/11	145	65.5	5	84	94/19	160	49	5	84	151.5	60	5	84
Mean	11/12	140.2	56.9	8.0	33.9	–	149.9	49.4	10.3	20.7	141.6	58.7	4.5	131.9

* Site excluded from the calculation of the formation mean because $\alpha_{95} > 15°$.

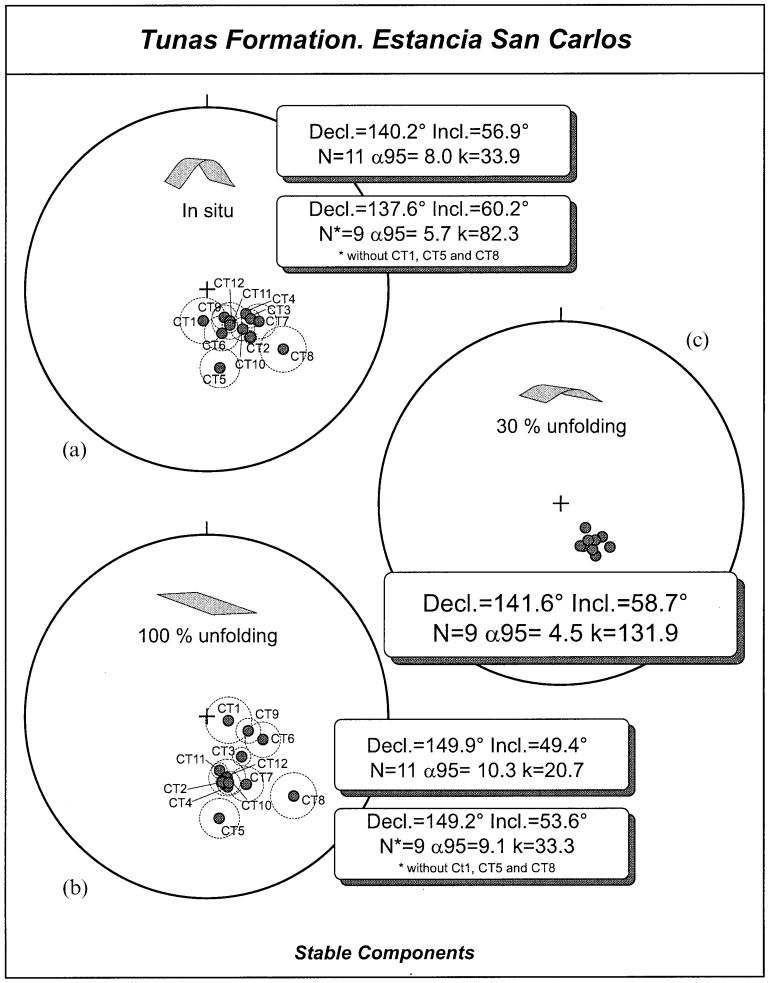


Figure 4. Tectonic tilt corrections for Population 1 (high-temperature characteristic remanence), in the Tunas Formation, estancia San Carlos. (a) *In situ*; (b) full 100 per cent tilt-corrected; and (c) partially (30 per cent) tilt-corrected. Statistics are shown for all sites (upper rectangle) and excluding sites in the fold hinges (lower rectangle) in (a) and (b). CT1 was excluded from the calculation of the formation mean because $\alpha_{95} > 15°$. See also Table 1 and Fig. 5.

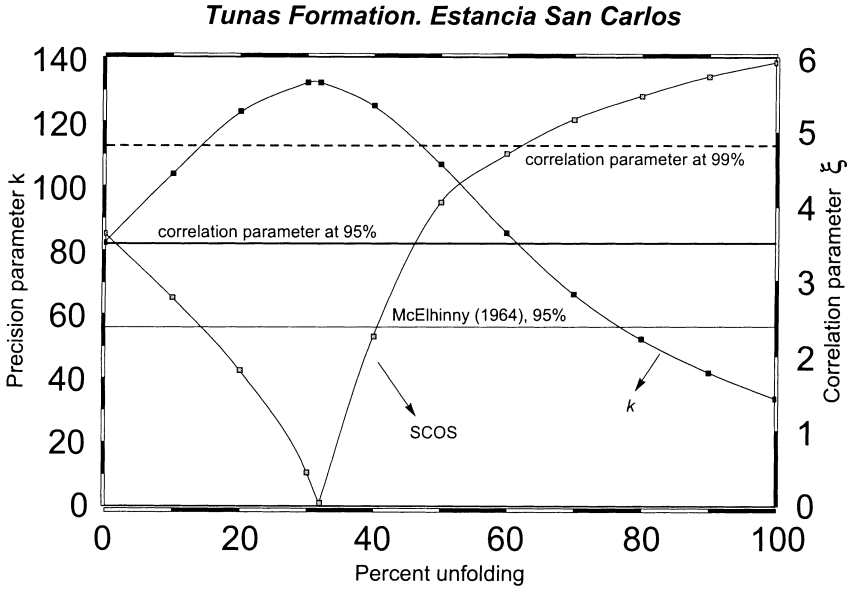


Figure 5. Plot of the statistical parameter k (Fisher's precision parameter; Fisher 1953) and ξ (SCOS: correlation parameter, McFadden 1990a), for the site mean direction of the Tunas Formation in the estancia San Carlos, versus percentage of tectonic correction. Lines are significance levels: 99 per cent and 95 per cent according to McFadden's (1990b) and McElhinny's (1964) fold test, respectively. See also Table 1 and Fig. 4.

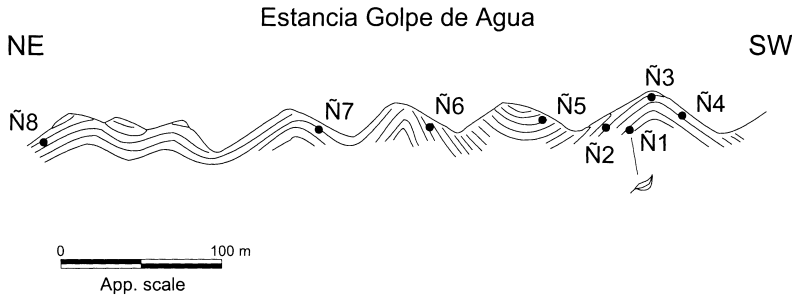


Figure 6. Schematic structural profile of the Tunas Formation in estancia Golpe de Agua, showing the positions of the sampling sites.

630 °C and 730 °C, and either a linear decay towards the origin or an abrupt decay of a very stable remanence (Fig. 7). For a few samples, particularly those from sites N4 and N6, the demagnetization paths were either curved (Fig. 7d) or linear. These sites have a slightly different lithology from the others, and are characterized by a greyish sandstone that is slightly coarser than the sandstones at the other sites. At sites N2 and N8, one and two samples, respectively, were discarded because of obscure markings on the orientated blocks.

The high-temperature characteristic remanence directions from the Tunas Formation in the estancia Golpe de Agua were grouped as Population 1 (Table 2). All sites showed positive (downwards) inclinations and good within-site grouping of directions ($\alpha_{95} < 15^\circ$ and $k > 20$). The high unblocking temperatures suggest that haematite is the main magnetic carrier.

A low-temperature component (below 500 °C) was found in some samples and grouped as Population 2. These components have negative (upwards) inclinations and northward declinations.

Analysis

The group mean direction *in situ* for Population 1 in estancia Golpe de Agua (Fig. 8), with $N = 8$, is Declination = 147.1°, Inclination = 51.3°, $\alpha_{95} = 15.2^\circ$, $k = 14.2$. Full 100 per cent bedding correction decreases the dispersion, yielding a mean direction of Declination = 143.1°, Inclination = 52.1°, $\alpha_{95} = 9.7^\circ$; $k = 33.6$, $N = 8$. Application of stepwise structural correction shows a clear improvement in the grouping of directions (Figs 8 and 9) at around 70 per cent of bedding correction: Declination = 145.5°, Inclination = 53.7°, $\alpha_{95} = 8.9^\circ$; $k = 47.4$, $N = 7$. The minimum correlation parameter of McFadden (1990b) is at 66 per cent ($\xi_{\text{Minimum}} = 0.260$), excluding site N3 because it is located in the fold hinge (McFadden 1990a). The fold test indicates a syntectonic magnetization at the 99 per cent confidence level. The magnetization of the Tunas Formation in the estancia Golpe de Agua was acquired during folding.

Population 2 shows more dispersion and was isolated in less than half of the specimens between room temperature

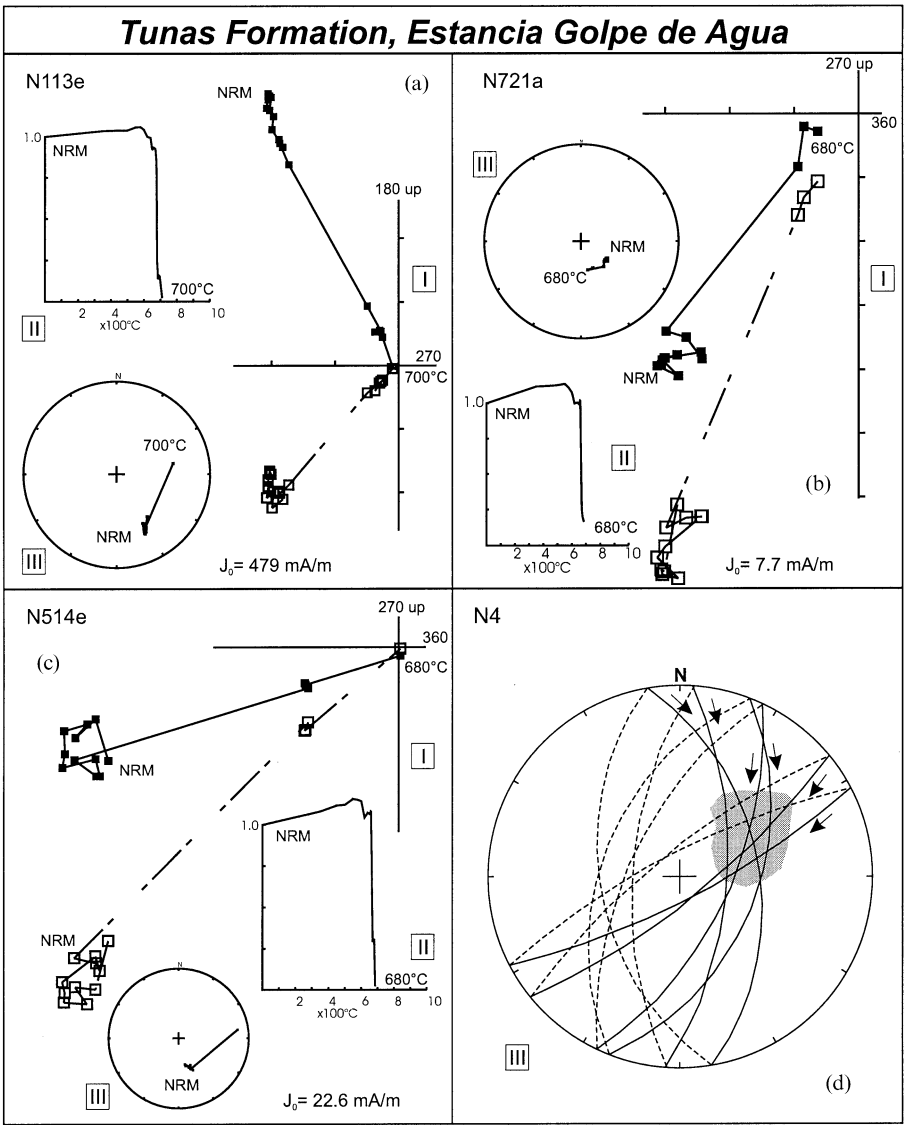


Figure 7. Representative thermal demagnetization results of some specimens from the Tunas Formation, in the estancia Golpe de Agua (location shown in Fig. 1). I: Zijderveld diagrams. Open (filled) squares indicate representation in the vertical (horizontal) plane, in geographical coordinates. II: Demagnetization curves. III: Projection of an equal-area net. For (d), remagnetization circles are shown. The shaded area indicates the location of the mean direction.

Table 2. Mean high-temperature characteristic directions in the Tunas Formation in estancia Golpe de Agua. *N*/*n*: number of processed specimens/number of specimens used in the calculation of the mean; (*n*) number of great circles. Dec.: Declination (°); In.: Inclination (°); α_{95}(°) = semiangle of the 95 per cent confidence cone; *k*: Fisher statistical parameter (Fisher 1953). Bedding: strike and dip (90° cw from given strike).

Means of Population 1

	in situ						with structural correction							
							100 per cent				30 per cent			
Site	N/*n*	Dec.°	In.°	α_{95}°	*k*	Bedding	Dec.°	In.°	α_{95}°	*k*	Dec.°	In.°	α_{95}°	*k*
N1	13/13	159	37	5	81	323/34	132	38	5	81	140	39	5	81
N2	13/13(3)	173	40	5	76.5	325/35.5	138	47	5	76.5	159.5	48	5	76.5
N3	12/12	138	48	6	56	48/4	138	45	6	56	–	–	–	–
N4	13/7(6)	91	58.5	14	25	137/32	157	67	14	25	134	68	14	25
N5	8/8	156	45	7	59	303/22	132	54	7	59	140.5	52	7	59
N6	14/9(5)	96	54	8	45.5	134.5/36.5	155.5	58	8	45.5	136	62.5	8	45.5
N7	13/13	148	55	4	118	137/24	147	51	4	118	168	49	4	118
N8	16/7	177	42	13	22	320/45	125	52	13	22	143	53	13	22
Mean	8/8	147.1	51.3	15.2	14.2	–	143.1	52.1	9.7	33.6	145.5	53.7	8.9	47.4

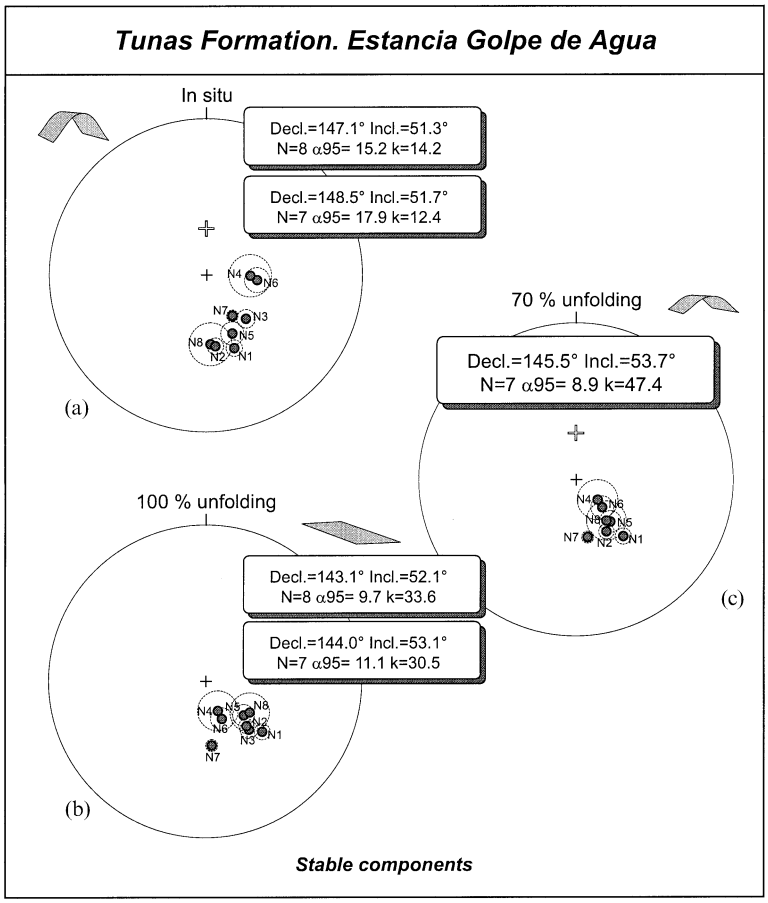


Figure 8. Tectonic tilt corrections for Population 1 (high-temperature characteristic remanence), in the Tunas Formation, estancia Golpe de Agua. (a) *In situ*; (b) full 100 per cent tilt-corrected; and (c) partially (70 per cent) tilt-corrected. Statistics are shown for all sites (upper rectangle) and excluding sites in the fold hinges (lower rectangle) in (a) and (b). See also Table 2 and Fig. 9.

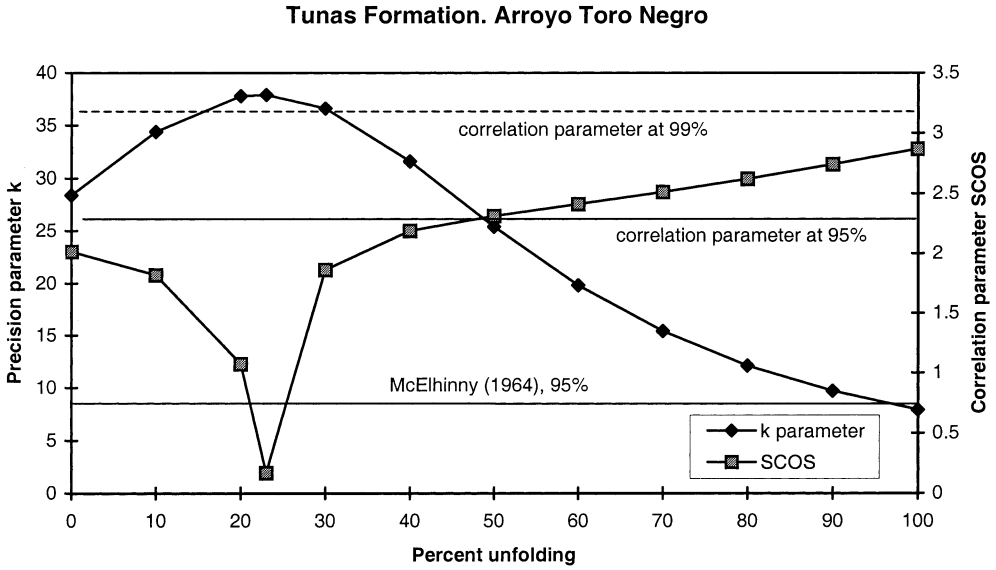


Figure 9. Plot of the statistical parameter k (Fisher's precision parameter; Fisher 1953) and ξ (SCOS: correlation parameter, McFadden 1990b), for the mean site direction of the Tunas Formation in the estancia Golpe de Agua, versus percentage of tectonic correction. Lines are significance levels: 99 per cent and 95 per cent according to McFadden's (1990b) and McElhinny's (1964) fold test, respectively. See also Table 2 and Fig. 8.

and 500 °C. Within sites, some directions appear random, and others are grouped close to a declination of 0°. This direction *in situ* resembles the present geomagnetic field direction, and appears to be a secondary magnetization of recent origin.

ARROYO TORO NEGRO

This locality is situated between the Sierra de las Tunas and Sierra de Pillahuincó (Fig. 1). Five sites having different structural and stratigraphic positions were sampled, with at least five cores at each. Two specimens from each core (a total of 42 specimens) were submitted to stepwise thermal demagnetization. NRM intensities range from 1 to 10 mA m^{-1} in the red beds. Magnetic behaviour was similar in all of the demagnetized specimens, yielding a very stable single-component magnetization with unblocking temperatures between 620 °C and 680 °C (Fig. 10).

The well-defined stable characteristic remanence directions for the Tunas Formation of the arroyo Toro Negro locality were grouped into Population 1 (Table 3), with positive (downwards) inclinations and good within-site consistency ($\alpha_{95} < 15°$ and $k > 20$). The high unblocking temperatures suggest haematite as the main magnetic carrier. In some specimens, a second component was isolated, with unblocking temperatures under 500 °C, negative inclinations (upwards), and random declinations.

Analysis

The *in situ* group mean direction of Population 1 (Fig. 11) is Declination = 158.9°, Inclination = 63.8°, $\alpha_{95} = 15.7°$; $k = 24.6$, $N = 5$. Full 100 per cent bedding correction increased the dispersion: Declination = 182.6°, Inclination = 52.0°, $\alpha_{95} = 27.2°$ and $k = 8.9$, $N = 5$ (Fig. 11).

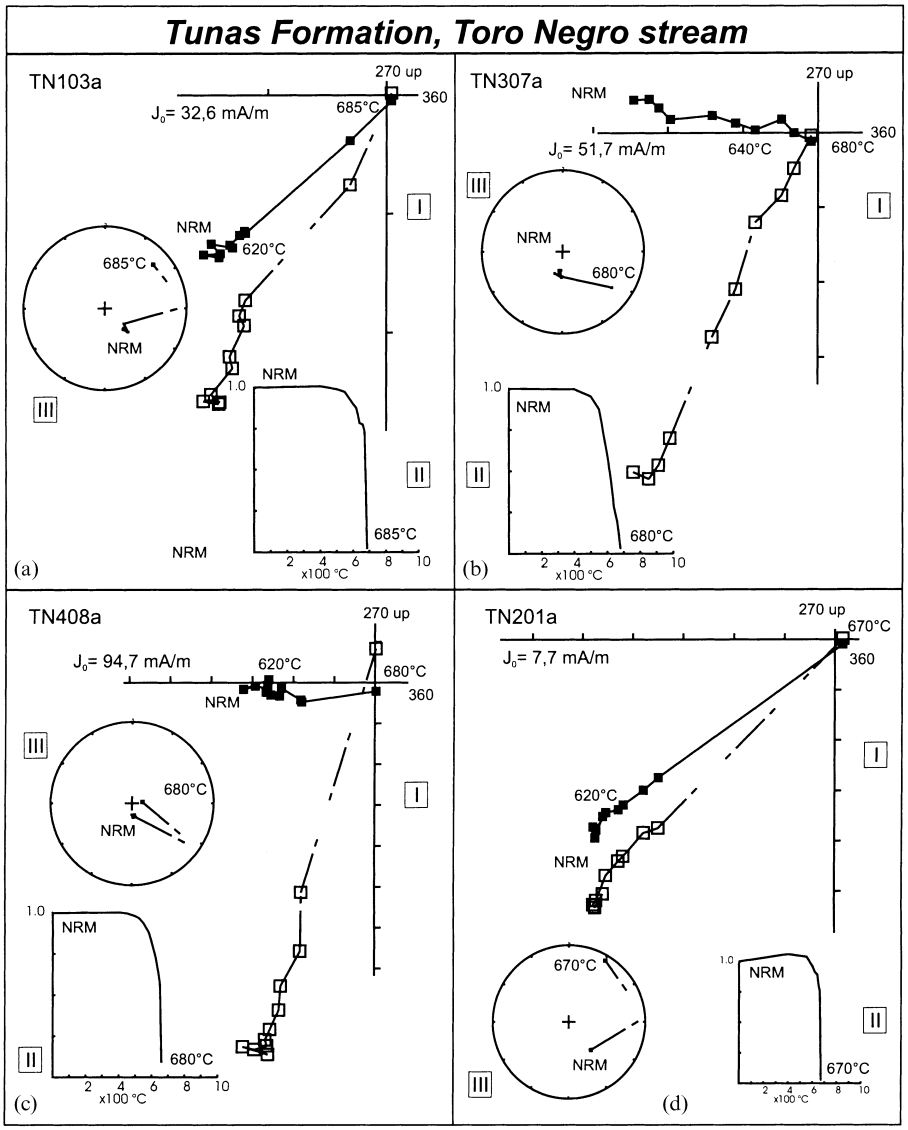


Figure 10. Representative thermal demagnetization results of some specimens from the Tunas Formation, in the arroyo Toro Negro (location shown in Fig. 1). I: Zijderveld diagrams. Open (filled) squares indicate representation in the vertical (horizontal) plane, in geographical coordinates. II: Demagnetization curves. III: Projection of an equal-area net.

Table 3. Mean high-temperature characteristic directions in the Tunas Formation in the arroyo Toro Negro. *N*/*n*: number of processed specimens/number of specimens used in the calculation of the mean; Dec.: Declination (°); In.: Inclination (°); α_{95}(°) = semiangle of the 95 per cent confidence cone; *k*: Fisher statistical parameter (Fisher 1953). Bedding: strike and dip (90° cw from given strike).

Means of Population 1

	in situ						with structural correction							
							100 per cent				30 per cent			
Site	N/*n*	Dec.°	In.°	α_{95}°	*k*	Bedding	Dec.°	In.°	α_{95}°	*k*	Dec.°	In.°	α_{95}°	*k*
TN1	6/6	122	56.5	6.5	105	140/40	178	48	6.5	105	137	58	6.5	105
TN2	8/8	160	46.5	9.5	35	50/05	158	42	9.5	35	–	–	–	–
TN3	2/2	187.5	65	–	–	320/31	105	69.5	–	–	175	69.5	–	–
TN4	5/5	191	70	6	179	141/40	216	33	6	179	203	62	6	179
TN5	5/5	147.5	70	7.5	104	141/40	202	43.5	7.5	104	170	67	7.5	104
Mean	5/5	158.9	63.8	15.7	24.6	–	182.6	52.0	27.2	8.9	169.6	66.4	15.1	37.9

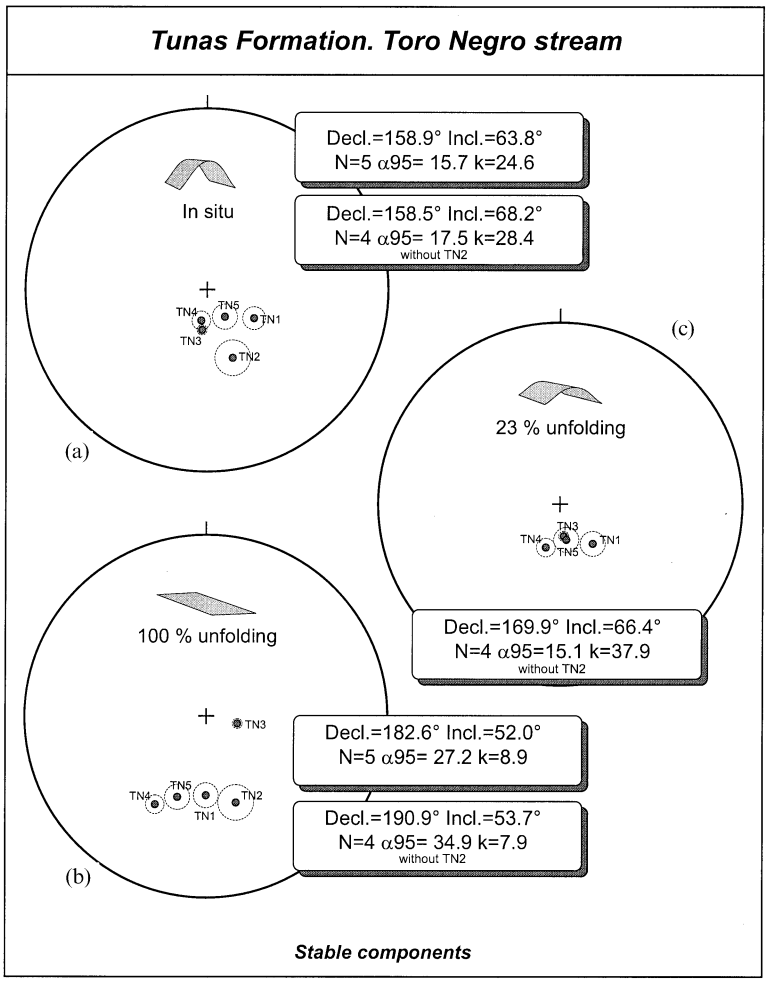


Figure 11. Tectonic tilt corrections for Population 1 (high-temperature characteristic remanence), in the Tunas Formation, arroyo Toro Negro. (a) *In situ*; (b) full 100 per cent tilt-corrected; and (c) partially (40 per cent) tilt-corrected. Statistics are shown for all sites (upper rectangle) and excluding sites in the fold hinges (lower rectangle) in (a) and (b). See also Table 3 and Fig. 12.

Application of stepwise structural correction shows the best grouping of directions (Figs 11 and 12) near 23 per cent of bedding correction: Declination = 169.6°, Inclination = 66.4°, $\alpha_{95} = 15.1°$; $k = 37.9$, $N = 4$. Site TN2 was excluded because it is located in the fold hinge (McFadden 1990a). Application of definition 1 of McFadden's (1990b) fold test indicates that this magnetization is syntectonic, with a minimum parameter correlation ($\xi_{\text{Minimum}} = 0.170$) at 23 per cent bedding correction,

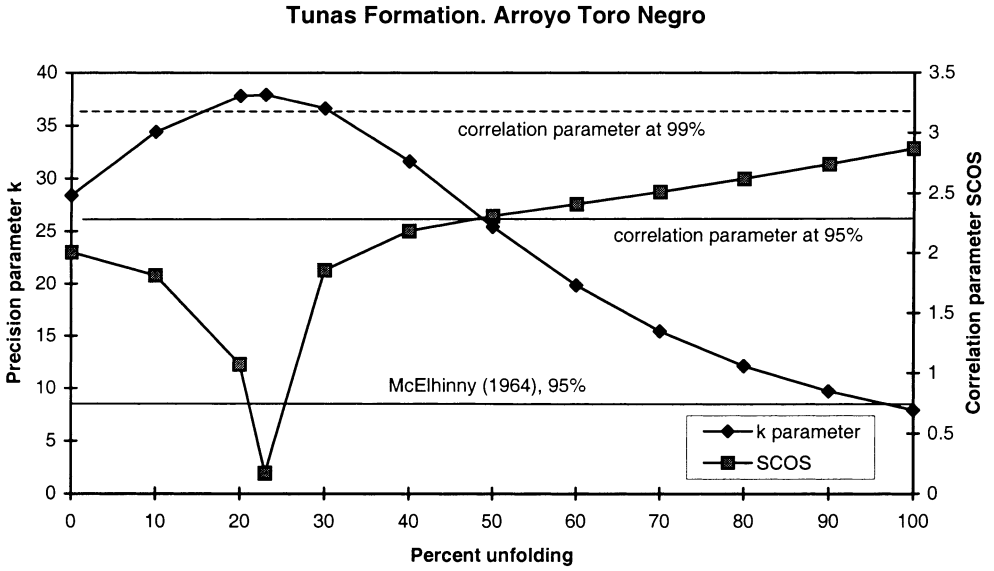


Figure 12. Plot of the statistical parameter k (Fisher's precision parameter; Fisher 1953) and ξ (SCOS: correlation parameter, McFadden 1990b) for the mean site direction, of the Tunas Formation in the Toro Negro stream, versus percentage of tectonic correction. Lines are significance levels: 99 per cent and 95 per cent according to McFadden's (1990b) and McElhinny's (1964) fold test, respectively. See also Table 3 and Fig. 11.

although it cannot be statistically distinguished from post-tectonic at the 99 per cent confidence level as a consequence of the small sample size.

These data suggest that the magnetization of the Tunas Formation of the arroyo Toro Negro locality, like those at estancia San Carlos and Golpe de Agua, was acquired during or after the folding of the sequence.

DISCUSSION AND TECTONIC IMPLICATIONS

Results presented here support the interpretation that the rocks of the Tunas Formation in the Sierra de las Tunas have a characteristic stable remanent magnetization carried by haematite, with exclusively reverse polarity consistent with magnetization assigned to the Kiaman interval.

Geological evidence of quasi-syntectonic deposition for the Tunas Formation is available. Growth folds at the arroyo Toro Negro locality were described by Rossello *et al.* (1993) and by López Gamundi *et al.* (1995). They are indicative of deformation contemporaneous with deposition. These authors note also that sediment transport was in the same direction as the tectonic transport.

A formation mean based on all accepted sites for the Tunas Formation is Declination = 147.4°, Inclination = 58.8°, $\alpha_{95} = 4.8°$; $k = 47.3$, $N = 20$, indicating a syntectonic magnetization at the 95 and 99 per cent levels of confidence (Fig. 13). We have calculated the palaeopole positions for each locality, based on the appropriate unfolding tests (Table 4).

From the virtual geomagnetic poles for each site, a syntectonic palaeomagnetic pole called *Tunas* was calculated: Latitude: 63.0°S; Longitude: 13.9°E, $A_{95} = 5.4°$ and $K = 39.7$, $N = 19$ (Table 4). This mean excluded the virtual geomagnetic pole site TN4 (which lies outside a window of radius 40° about the mean). The positions of poles T1, T2 and T3 are shown in Fig. 14, which also lists selected late Palaeozoic and early

Table 4. Palaeomagnetic poles calculated for each locality: San Carlos (T1), Golpe de Agua (T2) and Toro Negro (T3), and the palaeomagnetic pole for the Tunas Formation in the Sierra de las Tunas. Locality; Plat: pole latitude (°S); Plong: pole longitude (°E); α_{95} (°) = semiangle of the 95 per cent confidence cone; k: Fisher statistical parameter (Fisher 1953); n: number of virtual geomagnetic poles used for the calculation of the pole.

Locality	Plat	Plong	α_{95}°	k	n
Ea. San Carlos (T1)	61.0°S	13.0°E	6	74	9
Ea. Golpe de Agua (T2)	62.0°S	22.9°E	10.3	35.5	7
Ayo. Toro Negro (T3)	77°S	327.9°E	23.5	16	4
Tunas Pole	63.0°S	13.9°E	5.4	39.7	20

Mesozoic poles from South America. The new pole is coincident with other early Permian poles. The palaeofloristic and palaeofaunistic associations found in the Bonete and Tunas formations (Harrington 1947; Archangelsky & Cúneo 1984) indicate an early Permian age for these rocks (Asselian to Artinskian). As mentioned above, however, the magnetization found in these rocks does not correspond to the age of deposition but to the time of folding and deformation. This suggests that deformation occurred very soon after the deposition of the Tunas Formation. Andreis *et al.* (1979) concluded that the haematite of the Tunas Formation has a primary origin (*detrital* or *early diagenetic*). It appears likely that deformation was diachronous, and began in the western areas of the Tunas depositional basin while deposition was taking place in eastern areas.

The early Permian deformation responsible for the magnetization described here may correspond to the activity of the San Rafaelic orogenic phase (Azcuy & Caminos 1987; Ramos 1988). This phase has been recognized mainly in the western areas of Argentina and has been linked to remagnetization on a regional scale (Rapalini & Tarling 1993). According to Llambías & Sato (1995), the beginning of the San Rafaelic

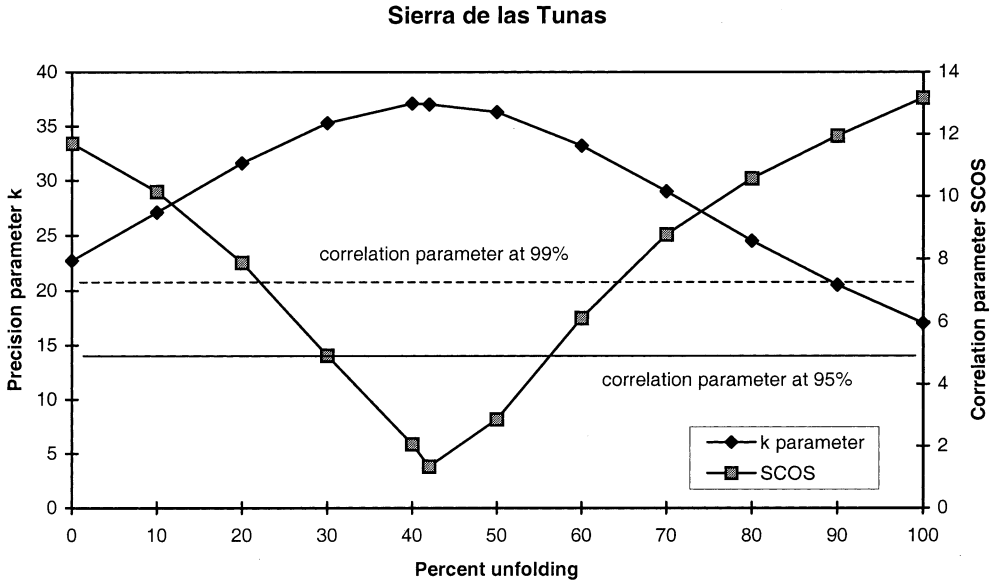
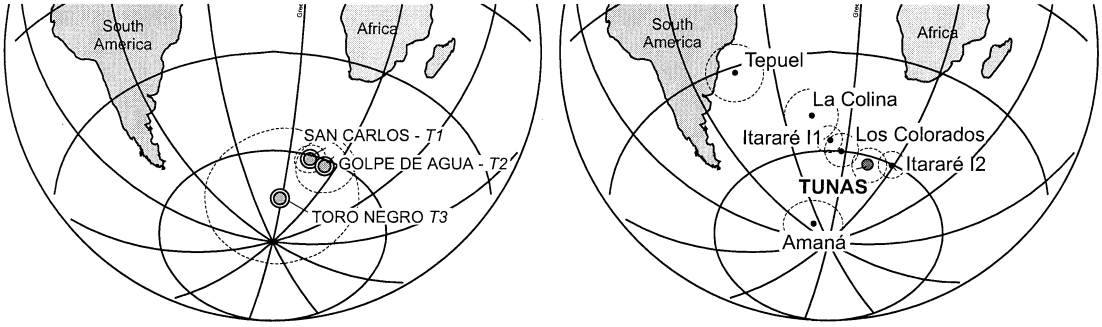


Figure 13. Plot of the statistical parameter k (Fisher's precision parameter; Fisher 1953) and ξ (SCOS: correlation parameter, McFadden 1990b), for the mean site direction of the Tunas Formation computed for the syntectonic magnetization, versus percentage of tectonic correction. Lines are significance levels: 99 per cent and 95 per cent according to McFadden's (1990b) fold test.



PALEOMAGNETIC POLE	α95°	LAT.°	LONG.°	PLAT.°	PLONG.°	AGE	REFERENCES
AMANA Ar	8	-30.3	292.3	-83	317	248.5-Ti	Valencio *et al.*, 1977
ITARARE I2 Br	4	-21.5	312.8	-60.3	29.5	280.5-Ps	Pascholati 1983
Lw. COLORADOS Ar	5	-29.5	293	-60	358	293-Pi	Embleton 1970
TUNAS Ar	5.1	-38	298.2	-62.7	16.6	290-Pi	This work
LA COLINA Ar	8	-30	293	-49	343	295-Pi	Sinito *et al.*, 1979
ITARARE I1 Br	4	-21.5	312.8	-56.7	350.6	293-Pi	Pascholati 1983
TEPUEL Ar	8.5	-43.5	289.6	-31.7	316	318-Cs	Rapalini *et al.*, 1994

Figure 14. Positions of the T1, T2, T3 and mean Tunas palaeomagnetic poles computed from the syntectonic magnetization of the Tunas Formation in San Carlos, Golpe de Agua and Toro Negro localities, and their 95 per cent confidence ovals. Selected late Palaeozoic and early Mesozoic South American poles from the cratonic areas are also shown.

orogenic phase in the Cordillera Frontal was during the Asselian (earliest Permian). According to the numerical time-scale for the Permo–Triassic (Menning 1995), it is possible to place the Asselian between 296 and 290 Ma. On the other hand, the age assigned to the Tunas Formation based on fossil content is from Asselian to Artinskian; that is, between 296 and 277 Ma. The foregoing implies an age for the Tunas palaeopole of approximately 296–290 Ma.

The palaeopolar positions of T1 and T2 are almost coincident, which is logical as the localities are close to each other (see Fig. 1). However the deformation was not perfectly synchronous. In estancia San Carlos the magnetization is syntectonic at 30 per cent of unfolding, while in Golpe de Agua (located to the east at a higher stratigraphic position), the magnetization was isolated at 70 per cent of unfolding. This could speculatively be interpreted as a migration of the deformation front to the ENE.

Unpublished geological and palaeomagnetic results from the Sierra de Pillahuincó, further east in the Sierras Australes, suggest a more complex regional history of deformation and magnetization (Tomezzoli 1997). In the Sierra de Pillahuincó, the upper part of the Tunas Formation, the magnetization is both younger and pretectonic, supporting the concept of an advance of the orogenic front to the ENE.

CONCLUSIONS

A palaeomagnetic study of the Tunas Formation in the Sierras Australes of Buenos Aires province shows that rocks from the San Carlos, Golpe de Agua and arroyo Toro Negro (situated in the Sierra de las Tunas) localities are carriers of a syntectonic magnetization acquired during the folding of the sequence. This is thought to correspond in time to the San Rafaelic orogenic phase of the Early Permian. It is possible to constrain the age of the Tunas magnetization, and thus of its palaeomagnetic pole, by biostratigraphic control, regional correlation and position in the APWP, to the early Permian, within the approximate range of 296–290 Ma. Deformation in the late Palaeozoic occurred very soon after the deposition of the lower part of the Tunas sequence. It seems likely that deformation began earlier in the western areas of the Tunas depositional basin, and was contemporaneous with deposition in the eastern areas, consistent with the ENE advance of the orogenic front. Considering that the collision between Patagonia and the South American craton began during Devonian times, this early Permian deformation may represent the final consolidation between these continental blocks.

ACKNOWLEDGMENTS

This palaeomagnetic study is part of the doctoral dissertation of RNT, supported by the Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET-Argentina) and the Universidad de Buenos Aires (Proyecto UBACYT-EX135), Laboratorio de Palaeomagnetismo 'D.A. Valencio', Departamento de Ciencias Geológicas. We recognize special debts of gratitude to Dr E. Cristallini, for his invaluable collaboration during all phases of this work, and to Drs A. Rapalini and H. Vizán for continuous help and critical comment. Drs S. Japas, J. Sellés Martínez and J. Parés collaborated with the field studies. Drs C. Mac Niocaill and W.D. MacDonald offered helpful suggestions which greatly improved this contribution. Two anonymous reviewers gave constructive suggestions on the original manuscript. The administrators at estancias San Carlos and Golpe de Agua kindly facilitated access to their properties for sampling.

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